

Decomposition:

Attack and take damage on both sides to control death and health. Display functions to display stats every turn.

Pattern Recognition:

For each turn, each character displays their Health, their name, if they have taken any damage, and for the zombie, the position they moved.

Abstraction:

The information each character needs is their health, name, how much damage they have taken, and if they are defeated when hp is equal to zero.

Algorithm design:

```
Start
>>
Declare Plant1 Object
>>
Declare Plant2 Object
>>
Declare Zombie Object
>>
Play turns Loop:
>> Plant Display
>> If plants die:
    >> Output "You lost you bum"
    >> Break loop
>> Zombie Display
>> Plants Attack
>> Zombie Move
>> if Zombie Distance == Nearest Plant:
    >> Zombie Attack nearest Plant
    >> If Plant HP = 0, dead
>> else Break Loop
```